

Micro-CFO: The Autonomous "Growth & Compliance" Engine for Indian MSMEs

Problem Statement

Indian MSMEs operate in a "Compliance Fog," navigating over 58,000 changing regulations (GST, MCA, Labour) while simultaneously missing out on millions in unclaimed government subsidies (PLI, Startup India). They lack the budget for a professional CFO, leading to a "double-loss" scenario: severe penalties for accidental non-compliance and lost "free money" from ignored government grants.

Motivation

- **The Gap:** Current Fintech is purely **transactional** (payment gateways) or **descriptive** (dashboards showing past history). There is no **prescriptive** tool that actively manages a business's future health.
- **The Opportunity:** To move beyond simple "chatbots" and build a **Multi-Agent GenAI System** that acts as a proactive partner—auditing bills via camera, negotiating payment terms via email, and hunting for subsidies automatically.

Application

- **Target Users:** Tier-2/3 Manufacturers, Proprietors, and Gig-Economy Freelancers.
- **Real-World Use Case:** A user takes a photo of a lunch bill and uploads it to the Micro-CFO WhatsApp bot.
 - **Immediate Action:** The bot warns, *"This looks like a personal expense. Do not claim GST or you risk an audit."*
 - **Strategic Action:** It then adds, *"By the way, since you purchased new machinery last week, you are eligible for the 'PMFME' subsidy scheme. Should I draft the application?"*

Proposed Method

We propose a Quad-Agent GenAI Architecture to handle distinct financial roles:

1. **Agent A: The Visual Auditor (Multimodal LLM):** Uses Vision-Language Models (like GPT-4o or Gemini Pro Vision) to scan user-uploaded photos of invoices/bills. It detects fraud and misclassification *before* data enters the ledger.
2. **Agent B: The Legislative Sentinel (RAG System):** A vector database continuously ingesting PDFs from the Ministry of Corporate Affairs (MCA) and RBI. It monitors real-time law changes and alerts the user only if it affects *their* specific business type.
3. **Agent C: The Subsidy Hunter (Semantic Matcher):** Performs semantic search across thousands of Government Scheme documents to match the user's transaction history with live subsidy criteria.
4. **Agent D: The Negotiator (Generative Writer):** A fine-tuned LLM that autonomously drafts polite emails to vendors asking for credit extensions or to clients chasing overdue invoices when cash flow is predicted to dip.

Datasets / Data Source

- **Legal & Govt Data:** Public PDFs of Indian Laws (GST Act, Companies Act) and Government Schemes (Startup India, MSME Ministry portal).
- **Financial Ground Truth:** Anonymized MSME bank statements and a dataset of "Compliant vs. Non-Compliant" invoices for training the Visual Auditor.
- **User Context:** Real-time financial data fetched via the **Account Aggregator (AA)** framework.

Experiments

- **The "Audit Challenge":** We will test the **Visual Auditor** against a dataset of 500 mixed invoices (faded, handwritten, non-compliant) to measure its False Acceptance Rate (FAR) for illegal GST claims.
- **Subsidy Recall Rate:** We will run the **Subsidy Hunter** against historical data of 50 real companies to see how many valid government schemes the AI could have retrospectively identified for them.
- **Negotiation A/B Testing:** Evaluate the success rate of AI-drafted "payment collection" messages vs. standard templates in securing early payments.

Novelty and Scope to Scale

- **Novelty:** "Micro-CFO" is the first solution to combine **Defense** (Audit/Compliance protection) with **Offense** (Subsidy hunting/Negotiation). It doesn't just save money; it *finds* money.
- **Scope to Scale:**
 - **Vernacular Scale:** The WhatsApp-first design allows deployment in Hindi, Tamil, and Telugu without UI changes.
 - **Sector Expansion:** The "Subsidy Hunter" module can be easily retrained for **Farmers (Kisan Credit)** or **Students (Scholarships)** by simply swapping the underlying PDF documents.